

asthma emergency department visits for all year, warm months and cold months (all excluding September and October due to fall peaks in asthma emergency department visits). Data on ED visits were obtained from the 11 New York City Health and Hospitals Corporation medical centers with emergency receiving facilities.

G.2.3.3 Mar and Koenig (2009)

Mar and Koenig (2009) evaluated the relationship between outdoor ozone in the summer and asthma aggravation. The authors used hospital data on daily asthma cases from 1998 to 2002 in Seattle with local monitored PM_{2.5} and ozone concentrations to assess the association between asthma visits to the emergency department and air pollutants. They analyzed 1-hour and 8-hour max ozone concentrations at 2 monitors in Greater Seattle. Asthma ED visits were analyzed at 0 through 5-day lags. The authors found that ozone exposure exacerbates asthma in people in the Seattle area, especially in children. Authors found that in adults during the warmer months between May and September, a 10 ppb increase in 8-hour maximum ozone concentration is associated with relative risk of asthma-related ED visits of 1.08 (1.02, 1.14) with a 4-day lag. In children, during the warmer months, a 10 ppb increase in 8-hour maximum ozone concentration is associated with relative risk of asthma-related ED visits of 1.11 (1.02, 1.21) with a 3-day lag. The difference in lag and relative risk between children and adults suggests that children are more immediately responsive to the adverse effects of ozone exposure.

G.2.3.4 Peel et al. (2005)

A number of emergency department studies have corroborated findings from mortality and hospital admission studies regarding an association of ambient air pollution and respiratory outcomes. More refined assessment has been limited by study size and available air quality data. Measurements of 5 pollutants: PM₁₀, ozone, NO₂, CO, and SO₂ were available for the entire study period (1 January 1993 to 31 August 2000); detailed measurements of particulate matter were available for 25 months. The authors obtained data on 4 million emergency department visits from 31 hospitals in Atlanta. Visits for asthma, chronic obstructive pulmonary disease, upper respiratory infection, and pneumonia were assessed in relation to air pollutants using Poisson generalized estimating equations. In single-pollutant models examining 3-day moving averages of pollutants (lags 0, 1, and 2): standard deviation increases of ozone, NO₂, CO, and PM₁₀ were associated with 1-3% increases in URI visits; a 2 microg/m increase of PM_{2.5} organic carbon was associated with a 3% increase in pneumonia visits; and standard deviation increases of NO₂ and CO were associated with 2-3% increases in chronic obstructive pulmonary disease visits. Positive associations persisted beyond 3 days for several of the outcomes, and over a week for asthma. The results of this study contribute to the evidence of an association of several correlated gaseous and particulate pollutants, including ozone, NO₂, CO, PM, and organic carbon, with specific respiratory conditions.

Emergency Room Visits, Asthma

The ozone coefficient and standard error are reported per 25 ppb increment of the maximum daily 8-hour average ozone level (Peel et al., 2003, Table 4). We used the results from the three cities combined. The relative risk is 1.022, with a 95 percent confidence interval of 0.996 to 1.049.

G.2.3.5 Sarnat et al. (2013)

The authors examined the influence of indoor air exchange rates as an effect modifier of associations between air pollutants and emergency department visits for asthma. The study analyzed associations between urban air pollutants (CO, NO_x, O₃, and PM_{2.5}) and asthma ED visits (ICD-9 codes 493 and 786.07) over a four-year period between 1999 and 2002 for residents of 186 zip codes in metropolitan Atlanta. A spatial surface of daily ambient air concentrations of pollutants was generated using AERMOD air quality modeling of 2002 NEI emissions combined with spatially interpolated background concentrations. AER was estimated at the zip code level based on an empirical model of direct and indirect predictors of AER collected from field surveys of building and meteorological characteristics. Effects of air pollution on asthma ED visits were estimated using a spatially resolved time series approach with Poisson generalized linear models, including stratified analyses for low or high AER areas. The authors found significant or near-significant positive effect modification of AER on the asthma ED visit effects of CO, NO_x, and PM_{2.5}. Ozone was a strong predictor of asthma ED visits but appeared unaffected by the AER. Overall, the study provides the first indication of short-term effect modification of air pollution risks with changes in AER. The HIF in BenMAP-CE is based on the association between ozone exposures and asthma ED visits for all AERs and all poverty levels (RR = 1.050, 95%CI 1.024-1.075) for a 26 ppb change in daily average ozone (3-day moving average of 0, 1, and 2 day lags). The beta value was adjusted to use the daily 8-hour max metric as described below.

G.2.3.6 Wilson et al. (2005)

Daily emergency room (ER) visits for all respiratory (ICD-9 460-519) and asthma (ICD-9 493) were compared with daily SO₂, ozone, and weather variables over the period 1998-2000 in Portland, Maine (population 248,000), and 1996-2000 in Manchester, New Hampshire (population 176,000). Seasonal variability was removed from all variables using nonparametric smoothed function (LOESS) of day of study. Generalized additive models were used to estimate the effect of elevated levels of pollutants on ER visits. Relative risks of pollutants were reported over their interquartile range (IQR, the 75th -25th percentile pollutant values). In Portland, an IQR increase in SO₂ was associated with a 5% (95% CI 2-7%) increase in all respiratory ER visits and a 6% (95% CI 1-12%) increase in asthma visits. An IQR increase in O₃ was associated with a 5% (95% CI 1-10%) increase in Portland asthmatic ER visits. No significant associations were found in Manchester, New Hampshire, possibly due to statistical limitations of analyzing a smaller population. The absence of statistical evidence for a relationship should not be used as evidence of no relationship. This analysis reveals that, on a daily basis, elevated SO₂ and O₃ have a significant impact on public health in Portland, Maine.

Emergency Room Visits, Asthma

The coefficient and standard error are taken from Wilson et al. (2005, Table 5).

G.2.4 Minor Effects

Table G-14 summarizes the additional health impacts functions used to estimate the relationship between ozone and minor effects. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table G-14. Additional Health Impact Functions for Ozone and Minor Effects

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
School Loss Days, All Cause	Chen et al.	2000	Washoe Co, NV	5-18	PM ₁₀ , CO	D8HourMax	0.015763	0.004985	Linear	All year, 8-hour max from 1-hour max.
School Loss Days, All Cause	Chen et al.	2000	Washoe Co, NV	5-18	PM ₁₀ , CO	D1HourMax	0.013247	0.004985	Linear	
School Loss Days, All Cause	Gilliland et al.	2001	Southern California	5-18		D8HourMax	0.007824	0.004445	Log-linear	All year, 8-hour max from 8-hour mean.
Minor Restricted Activity Days	Ostro and Rothschild	1989	Nationwide	18-64	PM _{2.5}	D8HourMax	0.002596	0.000776	Log-linear	8-hour max from 1-hour max.
Minor Restricted Activity Days	Ostro and Rothschild	1989	Nationwide	18-64	PM _{2.5}	D1HourMax	0.002200	0.000658	Log-linear	

G.2.4.1 Chen et al. (2000)

Chen et al. (2000) studied the association between air pollution and elementary school absenteeism (grades 1-6) in Washoe County, Nevada. Assuming that most children start kindergarten at age 5, the corresponding ages for grades 1 through 6 would be 6 through 11. Daily absence data were available for all elementary schools in the Washoe County School District. The authors regressed daily total absence rate on the three air pollutants, meteorological variables, and indicators for day of the week, month, and holidays. They reported statistically significant associations between both ozone and CO and daily total absence rate for grades one through six. PM₁₀ was negatively associated with absence rate, after adjustment for ozone, CO, and meteorological and temporal variables. The C-R function for ozone is based on the results from a multiple linear regression model with CO, ozone, and PM₁₀.

School Loss Days, All Cause

The coefficient and standard error are presented in Table 3 (Chen et al., 2000, p. 1008) for a unit ppm increase in the two-week average of daily one-hour maximum ozone concentration. This is converted to unit ppb increase by dividing by 1,000. The reported coefficient represents an absolute increase in absenteeism rate for a unit increase in ozone. If we apply this study to other locations, we assume that the same absolute increase will occur for a unit increase in ozone, regardless of the baseline rate. If the study location has a particularly high baseline rate, we may be overestimating decreases in absenteeism nationally, and vice-versa. As an example, consider if the baseline absenteeism rate were 10% in the study and 5% nationally. An absolute increase in absence rate of 2% associated with a given increase in ozone reflects a relative increase in absence rate of 20% for the study population. However, in the national estimate, we would assume the same absolute increase of 2%, but this would reflect a relative increase in the absenteeism rate of 40%.

An alternative approach is to estimate apply the relative increase in absenteeism rate in the C-R function by adjusting the results by the ratio of the national absenteeism rate to the study-specific rate. As a result, the percent increase in absenteeism rate associated with an increase in ozone is extrapolated nationally rather than the absolute increase in absenteeism rate. The incidence derivation section above describes the data used to estimate national and study-specific absence rates.

In addition to this scaling factor, there are two other scaling factors which are applied to the function. A scaling factor of 0.01 is used to convert the beta from a percentage (x 100) per unit increase of ozone to a proportion per unit increase of ozone. As a result it can be applied directly to the national population of school children ages 6 through 11 to estimate the number of absences avoided.

The final scaling factor adjusts for the number of school days in the ozone season. In the modeling program, the function is applied to every day in the ozone season (May 1 - September 30), however, in reality, school absences will be avoided only on school days. We assume that children are in school during weekdays for all of May, two weeks in June, one week in August, and all of September. This corresponds to approximately 2.75 months out of the 5 month season, resulting in an estimate of 39.3% of days ($2.75/5 * 5/7$). The C-R function parameters are shown below.

Population: population of children ages 6-11

Scaling Factor 1: Ratio of national school absence rate to study-specific school absence rate = 1.081. (National school absence rate of 5.5% obtained from the U.S. Department of Education (1996, Table 42-1). Study-specific school absence rate of 5.09% obtained from Chen et al. (2000, Table 1).)

Scaling Factor 2: Convert beta in percentage terms to a proportion = 0.01